

1 Introduction:

Clusters fundamentally alter the physics of electronic relaxation: an inner-valence hole that lies below the double-ionization threshold of its own monomer, can decay in a cluster by ionizing a neighbor. In Interatomic Coulombic Decay (ICD), the hole is filled by an outer-valence electron on the same monomer, and the excess energy is transferred to a neighbor through the Coulomb interaction, ejecting a low-energy secondary electron ^{1,2}. Since its prediction in 1997, ICD has been observed in rare-gas clusters, hydrogen-bonded systems, and biological systems ³. Low energy electrons efficiently induce strand breakage in DNA through dissociative attachment, making decay channels that exist only in the condensed phase directly relevant to radiation damage in tissue ⁴. An ongoing area of investigation is what happens after these decay events: the cascading channels and fragmentation pathways are difficult to disentangle experimentally.

Grieves and Orlando observed a high yield of H⁺(H₂O)_n from electron-bombarded water films deposited on argon multilayers, with a threshold near 29 eV ⁵. They assigned this to ICD based on the threshold matching the Ar 3s⁻¹ ionization energy and a fragment kinetic energy distribution consistent with two hole identities. However, calculations on the gas phase dimer reveal that this pathway is energetically forbidden: the dimer dicationic state is inaccessible due to the Coulomb repulsion between the final charges. A second channel exists in this regime that produces no free electron, in which the donor relaxes and transfers energy to a neighbor, leaving it neutral but electronically excited. Cederbaum termed this as virtual photon dissociation (VPD) ⁶ and derived a rate expression in which timescale is determined based on the acceptors photodissociation cross section.

In this work we determine ab initio the channel that operates in the argon-water dimer, identify the acceptor state, and compute the transfer rate by two independent routes. The acceptors turn out to be bound instead of dissociative, and furthermore we observe a bound excited manifold: of the fourteen states characterized here only one is dissociative along the O-H coordinate. Ab initio multiple spawning (AIMS) reveals the acceptor decaying within 4 fs, but after a period of 48 fs we did not reach the dissociative state. This contrasts with golden-rule predictions as based on the high photodissociation cross-section of water, VPD was assumed to be favored.

2 Methods:

All electronic structure calculations used OpenMolcas with aug-cc-pVTZ on oxygen and aug-cc-pVDZ on hydrogen. The water manifold was characterized using SA(20)-CASSCF(8,10) and oscillator strengths from RASSI, scanned along both O-H coordinates. The Ar 3s-1 vertical ionization energy was obtained at the XMS-CASPT2 level, giving 28.97 eV against an experimental 29.24 eV. Nonadiabatic dynamics used ab initio multiple spawning in PySpawn interfaced to OpenMolcas. Sixteen initial conditions were sampled from a Wigner distribution

built from the Hessian, constructed from the normal modes. Trajectories were launched onto the bright super excited VPD acceptor state with velocity Verlet at a 5 a.u. time step on SA(13)-CASSCF(8,10) surfaces to 48.4 fs. Running with this many states required a custom patching for the overlap parser in PySpawn. Fragmentation was assessed by classical propagation of 2000 trajectories on the computed one-dimensional surfaces at a 2 a.u. time step, with dissociation reported at 2.8 Å. These are lower bounds, as we are unable to describe the three-body fragmentation channel. The transfer coupling was evaluated using both a dipole-dipole matrix element from the transition dipole Ar 3p→3s, and from Cederbaum's golden-rule expression in which we use the measured photodissociation cross section.

3 Results and Discussion:

The vertical ionization energy of the Ar 3s orbital in the argon-water dimer was computed at the XMS-CASPT2 level, giving 28.97 eV, in agreement with the experimental value of 29.24 eV. Decay by ICD requires that the two-hole final state is lower in energy, calculated as the sum of the two monomer ionization potentials and the Coulomb repulsion between the two charges. The ICD final state therefore lies 3.5 eV above the resonance state, meaning ICD is closed for this geometry. Virtual photon dissociation, which has no Coulombic penalty, is thus the only decay pathway energetically accessible in this system. We note that ICD however is energetically allowed in the condensed phase, as calculations with EOM-CCSD demonstrate.

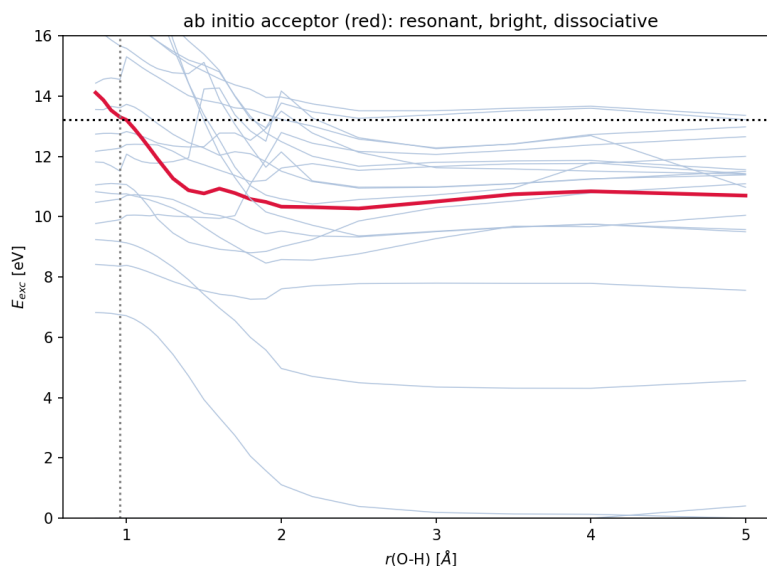


Figure 1: Excitation energies and oscillator strengths of the H₂O manifold at ground state geometry, SA(20)-CASSCF(8,10)/aug-cc-pVTZ. State 12 (red) at 13.303 eV with $f=0.157$ is the only state that is both resonant with the Ar 3s→3p transition (13.21 eV, dashed) and dipole-bright. Bands appear at expected positions, validating state tracking.

The acceptor state was characterized using SA(20)-CASSCF(8,10) as state 12, shown in red in **Figure 1**. Of the twenty states averaged, state 12 lies at 13.303 eV with an oscillator strength of 0.157 (RASSI), resonant with the donor to within 0.18 eV. Its oscillator strength is one of the largest in the resonant manifold. Along the dissociation coordinate, the excitation energy of the state decreases by 2.6 eV. State 12 is thus identified as an acceptor state, with the decrease of the excitation energy suggesting dissociative character.

Estimations of the coupling element of the energetic transfer between argon and water were obtained using the transition dipole moments with RASSI: the ground to acceptor excitation in water (0.6950 a.u.) and the argon 3p->3s transition (0.2319 a.u.). Using Equation 1, we obtain a coupling of 12.37 meV at the dimer equilibrium distance. This was cross validated using Cederbaum's golden-rule expression, showing a similar timescale of 1.20 ps at 3.5 Å and 380 fs at the contracted 2.9 Å. This agreement suggest that the transfer step is well described, which we will further discuss later.

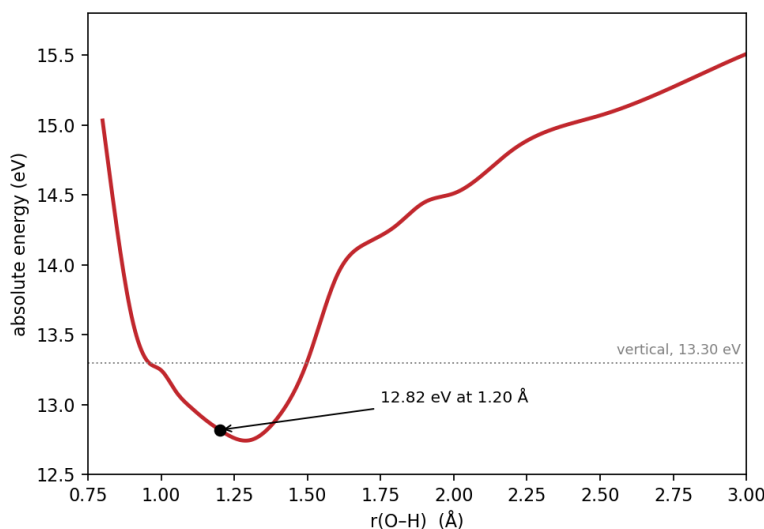


Figure 2: Absolute energy of the VPD acceptor along the $r(\text{O-H})$ coordinate, obtained from computed excitation energy and ground-state Morse potentials. The surface has a minimum near 1.20 Å (12.82 eV), and rises sharply afterwards. Classical propagation of Wigner-sampled trajectories on this surface gave only 0.1% dissociation.

As mentioned above, the excitation energy of the acceptor decreases by 2.6 eV between the equilibrium geometry and 2.0 Å, which would typically denote a dissociative state. To investigate this, 2000 initial-state conditions were sampled from a ground state distribution and propagated classically on the acceptor surface, with dissociation defined as the water bond exceeding 2.8 Å. Of these, only 0.1% dissociated, and we observed an average maximum bond length of 1.58 Å.

While this seems counterintuitive, the resolution lies in the absolute potential energy, which is shown in Figure 2. Excitation energy is the difference between the acceptor and the ground state,

but nuclei propagate on the absolute PES. Along the O-H coordinate the ground-state Morse potential rises by 5.1 eV while the excitation energy falls by 2.6 eV, an overall 2.5 eV increase. The absolute surface has a minimum near 1.2 Å, well into the bound regime. This behavior is consistent across the excited state manifold, with only one state allowing dissociation. Therefore, fragmentation in the argon-water dimer can only proceed nonadiabatically, with no direct route allowed.

To examine the decay of the bound acceptor state, we sampled sixteen initial condition and propagated using ab initio multiple spawning (AIMS) on SA(13)-CASSCF(8,10) surfaces launched from the acceptor. Trajectories reached 48.4 fs, with the states reported in Figure 3. We observe 260 spawning events, with the target-state distribution a downward funnel, propagating from state 12 to state 11, and only six total upward population transfers. The lowest state reached is state 3, after five femtoseconds, and after six no further spawning events occur. The acceptor decays rapidly through sequential internal conversions in a Rydberg-valence type manifold, physics that is inaccessible in the adiabatic representation. The states however remain bound, and the dissociative state is not populated within the window for this system.

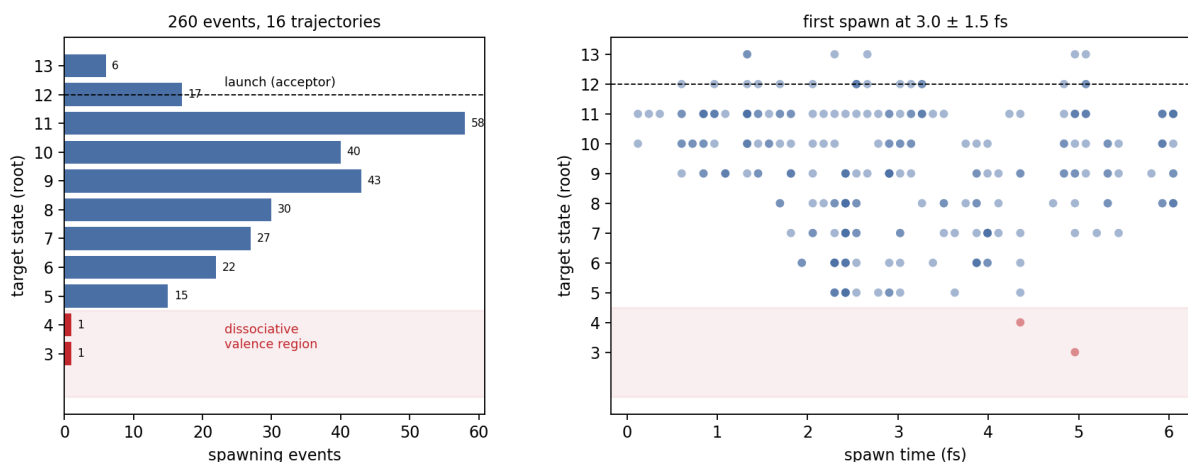


Figure 3: AIMS spawning events for the 16 Wigner-sampled trajectories launched on the VPD acceptor (root 12), SA(13)-CASSCF(8,10). Left: target state distribution, with 260 events. Right: spawning events against time, with red denoting the dissociative region.

Examination of decay pathways following a hypothetical transition to the dissociative state was done using classical methods, with the dissociation defined at the water bond exceeding 2.8 Å, and the hydrogen kinetic energy recorded at that crossing, a lower bound estimate. On the surface, 31.4% of the trajectories dissociate by 20 fs, with the hydrogen atom carrying 2.05 eV kinetic energy (Figure 4). The state thus fragments rapidly when populated, but whether the internal conversion cascade reaches this state lies beyond the current results. The reported kinetic energy interestingly is like the 1.7 eV kinetic energy of the $\text{H}^+(\text{H}_2\text{O})_4$ fragments of Grieves and Orlando, but the two observables are not directly comparable.

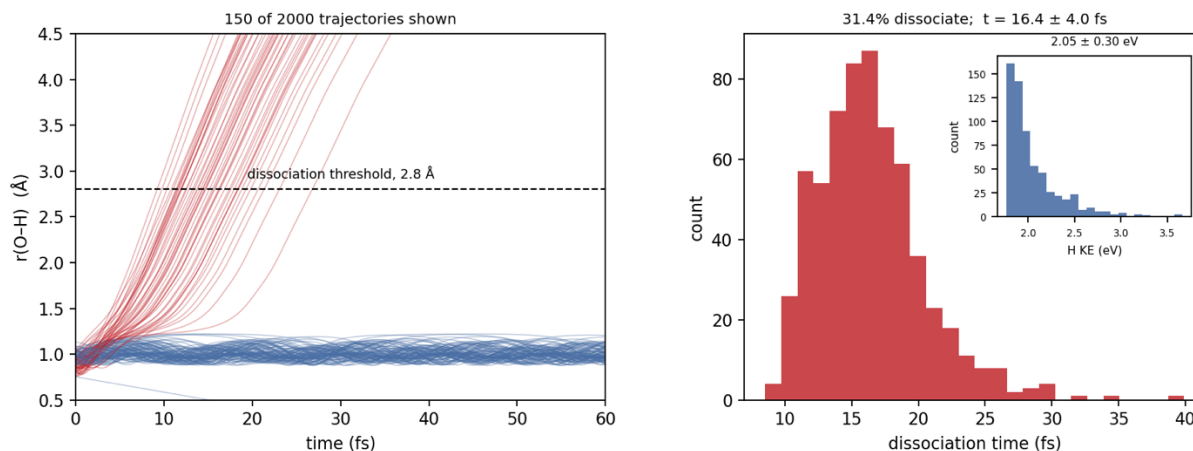


Figure 4: Classical propagation on the computed A surface, 2000 Wigner-sampled trajectories. Left: $r(\text{O-H})$ for 150 trajectories with red dissociating. Right: dissociation-time distribution, with inset the H-atom kinetic energy at the threshold. A is one state below the lowest state reached by AIMS cascade within the propagated window.

Conclusions and Future Outlook:

Grieves and Orlando's 29 eV threshold for water deposited on argon multilayers was assigned to Interatomic Coulombic Decay, but at the dimer equilibrium geometry this channel is closed energetically. Virtual photon dissociation (VPD) is the only open channel, with the accepting state the bright super-excited state of water with energy 13.30 eV, giving a predicted decay timescale of one picosecond. Our investigation however reveals that this state is bound, with classical propagation giving no dissociation, and only one of the fourteen lower states is dissociative along the O-H bond coordinate. AIMS shows the acceptor decaying within 3 fs and funneling into lower states, but not into the dissociative state in the 48 fs window. Further investigation requires greater inclusion of the condensed phase effects, but the density of states remains an issue.

In the future, a better surface with reduced state-averaging will allow us to extend the propagation window to determine if we reach the dissociative state. We initially ran energetics with 20 states averaging as we needed good description of states two through twelve, but the reduction on dynamics to 13 only shifted energies by 0.14 eV. The dissociation is perhaps under observed here as the classical fragmentation does not include the vibrational energy gained by the internal conversion cascade. Work on a model system with two identical acceptors, one that could undergo internal conversion could lead to better understand of the physics. We are currently attempting to run RT-CASCI with decoherent Ehrenfest on similar systems to gain insight into the hole population dynamics and fragmentation pathways.

Limitations:

Because of the scope of this project, we ran dynamics to 48 fs and did not observe a transition into the dissociative state. Fragmentation calculations use a rigid one-dimensional O-H coordinate with classical nuclei and cannot describe the three-body pathway. Finally, the dimer serves as a limiting case, we have shown that both ICD and ETMD(3) are allowed in the extended system.

References:

1. Cederbaum, L. S., J. Zobeley, and F. Tarantelli. "Giant Intermolecular Decay and Fragmentation of Clusters." *Physical Review Letters* 79, no. 24 (1997): 4778–81. <https://doi.org/10.1103/PhysRevLett.79.4778>.
2. Jahnke, Till, Uwe Hergenbahn, Bernd Winter, et al. "Interatomic and Intermolecular Coulombic Decay." *Chemical Reviews* 120, no. 20 (2020): 11295–369. <https://doi.org/10.1021/acs.chemrev.0c00106>.
3. Jahnke, T., H. Sann, T. Havermeier, et al. "Ultrafast Energy Transfer between Water Molecules." *Nature Physics* 6, no. 2 (2010): 139–42. <https://doi.org/10.1038/nphys1498>.
4. Alizadeh, Elahe, Thomas M. Orlando, and Léon Sanche. "Biomolecular Damage Induced by Ionizing Radiation: The Direct and Indirect Effects of Low-Energy Electrons on DNA." *Annual Review of Physical Chemistry* 66, no. 1 (2015): 379–98. <https://doi.org/10.1146/annurev-physchem-040513-103605>.
5. Grieves, Gregory A., and Thomas M. Orlando. "Intermolecular Coulomb Decay at Weakly Coupled Heterogeneous Interfaces." *Physical Review Letters* 107, no. 1 (2011): 016104. <https://doi.org/10.1103/PhysRevLett.107.016104>.
6. Cederbaum, Lorenz S. "Fragmentation of Molecules by Virtual Photons from Remote Neighbors." *The Journal of Physical Chemistry Letters* 11, no. 21 (2020): 8964–69. <https://doi.org/10.1021/acs.jpclett.0c02259>.

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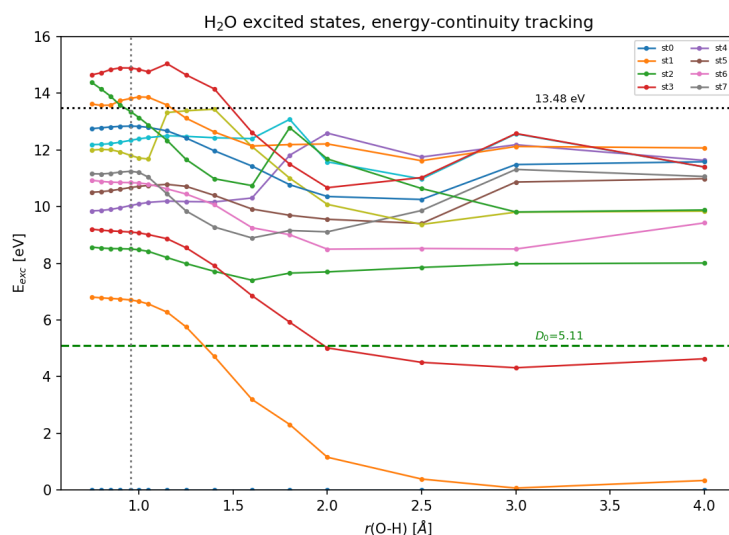


Figure S1: Tracked excitation energies of the water manifold along the O–H coordinate, SA-CASSCF(8,10)/aug-cc-pVTZ, assigned by energy continuity. The $\tilde{A}$ state (falling to 0.34 eV) and $\tilde{B}$ state (falling to 4.56 eV) are dissociative; the remaining states are bound or rising. State assignments in the dense region between 1.6 and 1.8 Å, where adjacent roots are separated by a few tenths of an eV, are less certain, but the $\tilde{A}$ and $\tilde{B}$ surfaces used quantitatively are unambiguous throughout.